WORK-STUDY FROM SEPTEMBER 2026 · EMBEDDED SYSTEMS, CONTROL & EV

Khalil LAGHA

M2 student — Intelligent Systems for Automotive & Aerospace (SIAA) · Université Paris-Saclay

Ranked 2nd of the M1 EEA cohort (15.43/20). Control (PID, LQR, Kalman, FOC), Matlab/Simulink validation, power electronics for electric vehicles — from the lab (GIPSA-lab) to the industrial field. Work-study rotation: 2 days company / 3 days university.

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- Authorized to work in France

01 Experience

Research Intern · GIPSA-lab, Grenoble — 04 – 07/2026 · 3.5 months

- Modeled a PEMFC fuel cell in Matlab/Simulink — specification → implementation → test approach, **6%** dynamic / **1%** static error.
- Tuned the regulation loop of a **boost DC-DC converter** (PI controller) — validated in simulation then on a real test bench.

Intern · SLB (Schlumberger) — 07/2025 · 1 month

- Wrote technical syntheses in English in an international industrial environment; **NEST & SIPP Level 2** HSE certifications.

Power Electronics & Automation Intern · EURL Lagha — 01/2024 · 15 d + 09/2024 · 1 month

- Worked on the electronics of real industrial equipment: a **buck DC-DC converter** (50 V → 0–48 V, **25 A**) and 4 transformer-rectifiers (**100 V DC / 80 A**, ripple < 5%).
- Contributed to the supervision HMI (**Siemens TIA Portal**, Ladder PLC); configured industrial control applications.
- Produced the electrical schematics in **EPLAN** — direct interface with manufacturing.

Power Grid Intern · Sonelgaz — 03/2024 · 15 d · observation

- Studied substation structure and the grid voltage chain (**400/220 kV → 400/230 V**); technical site visit.

02 Projects

Field-oriented control (FOC) — induction motor · Matlab/Simulink
Park transforms, PID controllers, observers, **Kalman filter**; cascaded current/speed loops — the core of electric-vehicle traction.

Neural network from scratch (MLP) · Matlab · open source (MIT)
Multilayer perceptron and backpropagation hand-coded, no toolbox — debugged, tested and published: github.com/KhalilEllahLAGHA/mlp-backpropagation-matlab.

Adjustable PWM generator · 555 timer + D flip-flops · Proteus
Adjustable frequency and duty cycle — simulated in Proteus then **built and fine-tuned on real hardware:** a complete electronics-diagnostics workflow.

Cascade control of a DC motor · Simulink · three-phase thyristor bridge
Outer speed loop, inner current loop; voltage control.

Work-study via CFA EVE
Université Paris-Saclay

2nd	15.43	C2
of the M1 EEA cohort (UGA)	M1 annual average /20	English

03 Education

M2 SIAA — Intelligent Systems for Automotive & Aerospace — 2026 –
Université Paris-Saclay (Évry) · work-study

M1 EEA (EECS) — 2025 – 26
Université Grenoble Alpes · 2nd in class

Engineering cycle, Electrical Engineering — 2021 – 25
École Nationale Polytechnique d'Alger · 4 of 5 years completed; prep classes: top-ranked

Baccalauréat in Mathematics — 2021
Highest honours — 16.93/20

04 Skills

Control: PID, LQR, Kalman · Embedded: C/C++, Arduino · Motor control (FOC) · EV power electronics · PLC / HMI / SCADA · AI: NN, GA, data fusion · Python (OOP) · Electrical CAD: EPLAN, AutoCAD

05 Software

Matlab/Simulink · Python · C/C++ · Arduino · Proteus · LTspice · Siemens TIA Portal · Step7 · EPLAN · AutoCAD Electrical · SolidWorks · Git/GitHub

06 Languages & certification

French	C1
English	C2
Arabic	native

SLB **NEST & SIPP Level 2** — 07/2025 (HSE)

07 Soft skills

Fast learner · Rigorous · Organized · Problem solving · Comfortable with tools

Portfolio : **khalilellahlagha.github.io/siaa/** · Code : **github.com/KhalilEllahLAGHA** · IEEE Student Branch ENP · VIC